

Experiment No-12

Locating Poles and Zeros in s-plane & z-plane

AIM: Write the program for locating poles and zeros and plotting pole-zero maps in s-plane and z-plane for the given transfer function.

Software Required: Matlab software

Theory:

Z-transforms

The Z-transform, like many other integral transforms, can be defined as either a *one-sided* or *two-sided* transform.

Bilateral Z-transform

The *bilateral* or *two-sided* Z-transform of a discrete-time signal $x[n]$ is the function $X(z)$ defined as

Unilateral Z-transform

Alternatively, in cases where $x[n]$ is defined only for $n \geq 0$, the *single-sided* or *unilateral* Z-transform is defined as

In signal processing, this definition is used when the signal is causal.

where $z = r.e^{j\omega}$

$$X(z) = \frac{P(z)}{Q(z)}$$

The roots of the equation $P(z) = 0$ correspond to the 'zeros' of $X(z)$

The roots of the equation $Q(z) = 0$ correspond to the 'poles' of $X(z)$

Example:

The zeros are: $\{-1\}$

$$H(z) = \frac{z+1}{(z-\frac{1}{2})(z+\frac{3}{4})}$$

The poles are: $\left\{ \frac{1}{2}, -\left(\frac{3}{4}\right) \right\}$

Program:

```

clc;
clear all;
close all;
%enter the numerator and denamenator coefficients in square brackets
num=input('enter numerator co-efficients');
den=input('enter denominator co-efficients');

% find poles and zeros
poles=roots(den)
zeros=roots(num)

% find transfer function H(s)
h=tf(num,den);

% plot the pole-zero map in s-plane
sgrid;
pzmap(h);
grid on;
title('locating poles and zeros on s-plane');

%plot the pole zero map in z-plane
figure
zplane(poles,zeros);
grid on;
title('locating poler and zeros on z-plane');

```

Result: Pole-zero maps are plotted in s-plane and z-plane for the given transfer function.

Output:

enter numerator co-efficients[1 -1 4 3.5]

enter denominator co-efficients[2 3 -2.5 6]

poles =

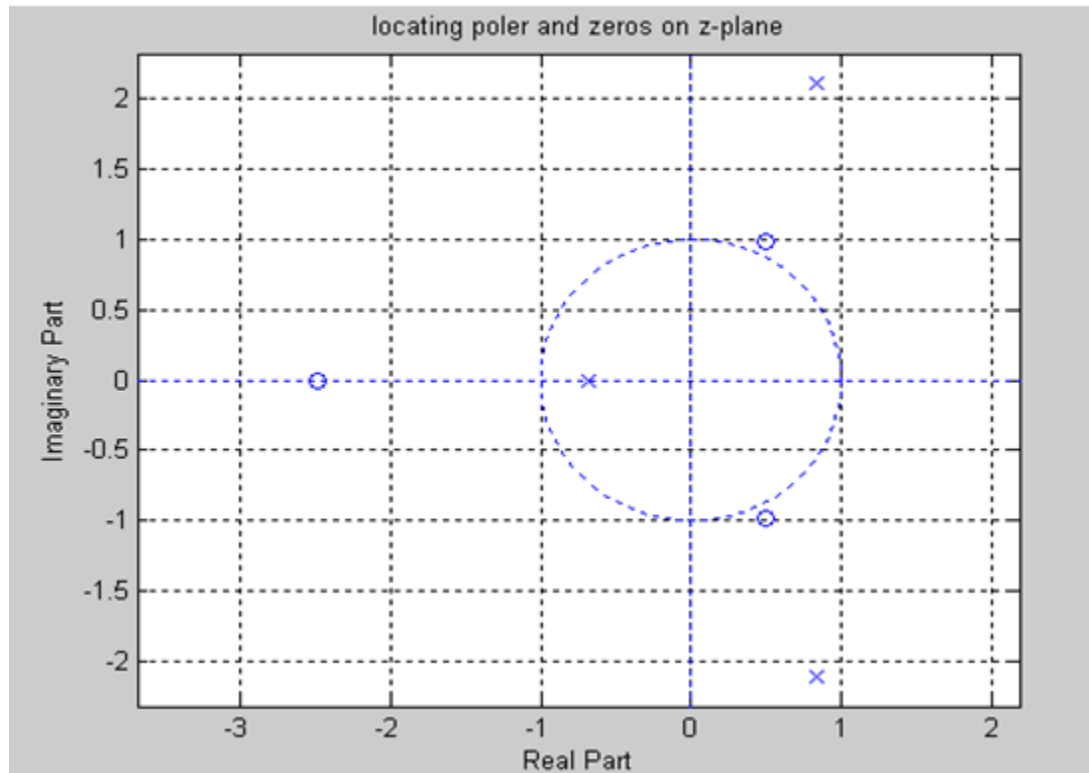
-2.4874

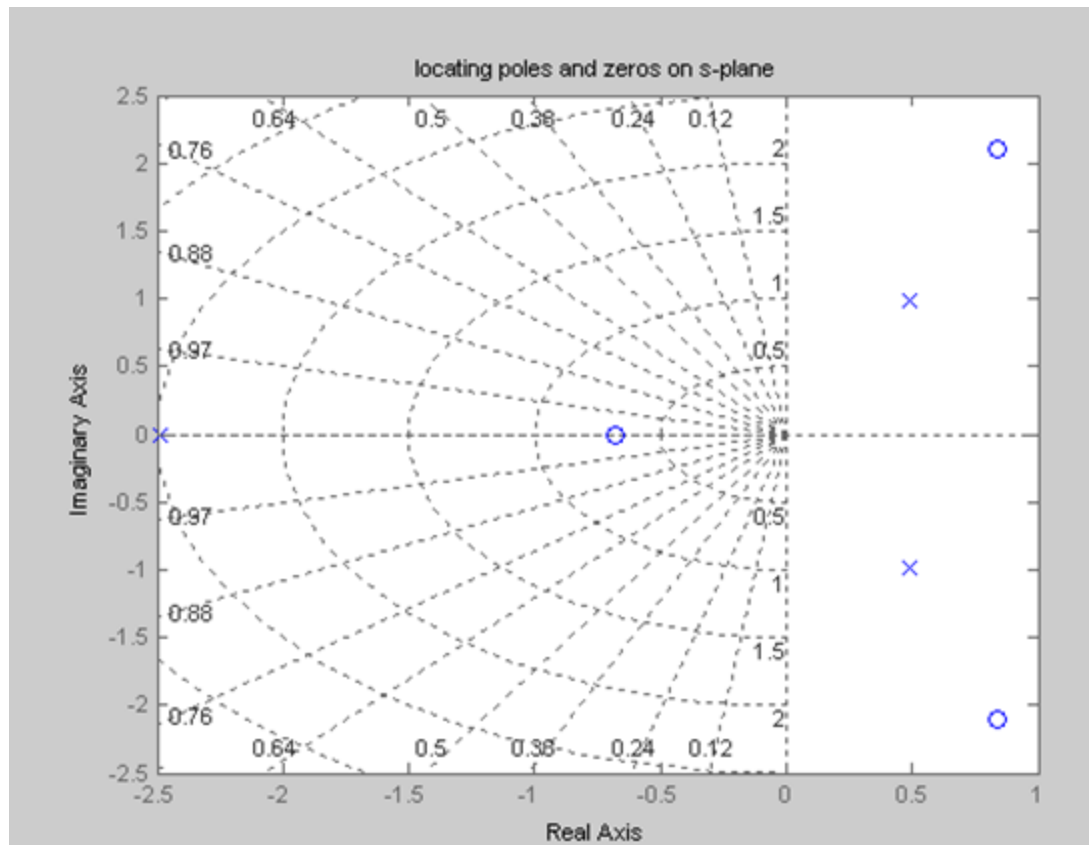
0.4937 + 0.9810i

0.4937 - 0.9810i

zeros =

0.8402 + 2.1065i

$0.8402 - 2.1065i$ -0.6805 



VIVA QUESTIONS:-

1. Study the details of pzmap() and zplane() functions?
2. What are poles and zeros?
3. How you specify the stability based on poles and zeros?
4. Define S-plane and Z-plane?
5. Define transfer function of the system?